

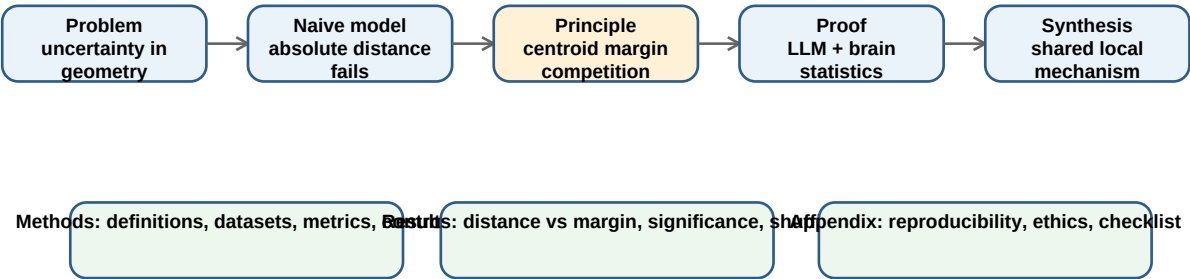
NeurIPS 2026 Pre-submission Checklist and Paper Structure Guide

Built for the current project: LLM emotion embedding geometry, centroid/margin-based ambiguity, and five-emotion brain fMRI geometry. This guide translates the provided NeurIPS reviewing guidelines, handbook requirements, and checklist guidance into concrete actions for the submission.

Decision item	Recommended choice for this paper	Why this is safest
Contribution type	General, with a clearly stated conceptual/mechanistic contribution	The work is empirical and conceptual rather than a pure theorem, use-inspired deployment, or standalone negative-result paper. It fits General while emphasizing a novel principle: uncertainty arises from relative geometric competition.
Core claim	Uncertainty/ambiguity is governed by centroid margin, not absolute distance	Supported by correlation, AUC, and shuffle controls. This is stronger than a descriptive brain-vs-LLM similarity claim.
Main risk	Overclaiming universality or biological equivalence	Avoid claiming that LLMs replicate the brain. Claim a shared local geometric mechanism tested across artificial and neural representations.
Main acceptance lever	Quality + clarity + originality	The project's strength is not SOTA performance, but a reproducible metric, strong controls, and a useful new framing of representation uncertainty.

1. Acceptance-oriented paper narrative

The paper should not be structured as a simple sequence of LLM analysis, then brain analysis, then comparison. That reads as exploratory. The stronger NeurIPS-facing structure is: define the problem, show why a natural explanation fails, introduce a better principle, validate it in LLMs, generalise it to brain data, then discuss supporting structure such as RDMs, valence-arousal alignment, and density.



Reviewer-facing logic: the paper should read as a mechanism paper, not only a descriptive LLM-vs-brain comparison.

Recommended title framing

Option	Title direction	Why it works
1	Centroid Margin Predicts Ambiguity in Artificial and Neural Emotion Representations	Direct, technically specific, and avoids overclaiming.
2	Uncertainty Emerges from Geometric Competition in Brain and Language Model Representations	Mechanism-first and broad enough for NeurIPS.
3	A Shared Local Geometry of Ambiguity Across Neural and Language Model Representations	Emphasizes cross-system contribution while staying cautious.

2. Final paper structure

Section	Purpose	Must contain	Reviewer criterion served
Abstract	One-paragraph argument for significance and originality.	Problem, failure of distance, margin principle, LLM + brain validation, key statistics, cautious scope.	Clarity, Significance, Originality
1. Introduction	Set up why uncertainty in representation geometry matters.	Representations are geometric; prior work shows structure/alignment; gap is how uncertainty emerges; contributions listed explicitly.	Clarity, Significance
2. Related Work	Show that you know the literatures and the novelty boundary.	LLM geometry, brain/RSA geometry, brain-LLM alignment, uncertainty/decision boundaries.	Originality, Quality
3. Problem Formulation	Define the mathematical objects before experiments.	x in R^d , labels, centroids, Euclidean distance, margin, uncertainty, RDM.	Clarity, Quality
4. Methods	Make the work reproducible.	LLM pipeline, brain fMRI pipeline, preprocessing, models, cross-validation, statistics, controls, compute.	Quality, Clarity, Reproducibility
5. Results I: Distance Fails	Show the natural explanation is insufficient.	Distance vs uncertainty: weak r and weak AUC. State why this matters.	Originality, Significance
6. Results II: Margin Works	Core empirical result.	Margin vs uncertainty: strong correlation, AUC, feature shuffle collapse. This is the main figure.	Quality, Significance
7. Results III: Cross-system Generalisation	Show the principle is not system-specific.	Repeat/compare margin behaviour in LLMs and brain; report matched metrics and controls.	Significance, Originality
8. Supporting Geometry	Explain broader representational structure without distracting from the main claim.	RDMs, centroid plots, VA analysis, density decay as supporting/negative insight.	Clarity, Completeness
9. Discussion	Convert results into insight.	Why margin captures competition, why density/distance are insufficient, what brain-LLM similarities/differences mean.	Significance, Originality
10. Limitations	Protect against reviewer objections.	5 emotions, centroid assumptions, dataset specificity, small centroid count for VA, no claim of universal brain equivalence.	Quality, Ethics
Conclusion	One clear takeaway.	Uncertainty is a relational property of representation geometry.	Clarity

3. Main-text figure checklist

The nine-page limit means every main figure must earn its space. The figure strategy should support the argument rather than merely display outputs.

Priority	Figure	Required panels / content	Main claim supported
Must-have Fig. 1	Mechanism overview	Schematic: points, centroids, distance, nearest competitor, margin. Include formula $\text{margin} = d_{\text{correct}} - d_{\text{nearest_other}}$.	Defines the new principle clearly.
Must-have Fig. 2	Distance fails, margin works	Panel A: distance vs uncertainty weak. Panel B: margin vs uncertainty strong. Panel C: AUC comparison. Panel D: feature-shuffle collapse.	Core technical proof.
Must-have Fig. 3	LLM pipeline result	Embedding centroids/clusters, LLM margin-uncertainty curve, model comparison if available.	Shows mechanism in artificial representations.
Must-have Fig. 4	Brain validation	LOSO decoding result, brain margin-uncertainty result, shuffle control.	Shows biological generalisation.

Priority	Figure	Required panels / content	Main claim supported
Optional Fig. 5	Global geometry support	RDM heatmap, centroid PCA, valence-arousal alignment.	Explains global structure and differences.
Appendix	Density decay	KDE/density plots and $p=0.46$ explanation.	Shows global density alone is insufficient.

4. Statistics checklist

Test	Status needed before submission	Report exactly	Why it matters
Distance vs uncertainty	Done for brain; repeat/confirm for LLM if central	Spearman r , p -value, Pearson r , AUC	Shows the naive radial explanation is weak.
Margin vs uncertainty	Core result	Spearman r , p -value, Pearson r , AUC	Shows ambiguity is relational/competitive.
Feature-shuffle control	Required	Shuffled r and p ; ideally plot real vs shuffled	Shows structure is not random.
LOSO brain decoding	Required for brain validity	Accuracy, balanced accuracy, chance, CI, permutation p	Shows brain space contains emotion signal.
Permutation tests	Required for central effects	Null mean, SD, p -value, number of permutations	Supports statistical significance.
Bootstrap CI	Strongly recommended	95% CI and resampling unit	Shows robustness and uncertainty of estimates.
RDM/RSA	Supporting	Distance metric, correlation method, whether shared labels are used	Prevents overinterpretation of cross-system geometry.
VA alignment	Supporting	Explained variance, correlation values, p -values, caveat $n=5$ centroids	Explains global axis structure.
Density decay	Supporting/negative result	p approx 0.46 and interpretation	Shows density alone is insufficient; strengthens margin mechanism.

5. Methods section requirements

The methods section must be detailed enough that an expert reader can reproduce the results. Do not assume reviewers know your notebooks.

Subsection	Minimum details to include
LLM data and embeddings	Dataset/source of emotion text examples; labels; number of samples per class; models used; embedding extraction layer/pooling; fine-tuning if used; train/test splits; preprocessing; random seeds.
Brain dataset	NeuroEmo/OpenNeuro ds005700 v1.2.0; 40 participants; five emotions; exclusion of control/noise blocks; fMRI BOLD definition; ROI extraction using Harvard-Oxford cortical atlas; final matrix 200 x 48; scaling/standardisation.
Geometry construction	Centroid formula; Euclidean distance; nearest competing centroid; RDM construction; whether metrics use original feature space or reduced space.
Uncertainty estimation	Classifier used; cross-validation protocol; probability calibration if relevant; uncertainty = $1 - P(\text{correct})$.
Margin metric	Formula; sign interpretation; why it captures competition; how it differs from raw distance.
Statistical tests	Spearman/Pearson; AUC definition; permutation test design; bootstrap unit; feature-shuffle control preserving feature marginals while destroying multivariate structure.
Implementation	Python packages, versions if available, MacBook Pro M4 Pro 24GB, CPU workflow, approximate runtimes.

6. Submission checklist mapped to NeurIPS checklist questions

Below is a draft answer strategy. The final paper should point to exact section numbers once written.

Checklist item	Recommended answer	What must be in paper/supplement
1. Claims	Yes	Abstract and Introduction state the exact scope: margin predicts uncertainty in tested LLM and brain emotion representations; no universal claim beyond evidence.
2. Limitations	Yes	Dedicated Limitations section: five emotion labels, centroid/prototype assumption, public fMRI dataset, limited brain dataset count, VA based on five centroids, LLM model coverage.
3. Theory/proofs	N/A or Yes-with-scope	If no theorem is claimed, answer N/A. If you include formal propositions, state assumptions and provide proof/sketch in appendix.
4. Experimental reproducibility	Yes	Provide pipeline, data sources, seeds, splits, metrics, code instructions or anonymized zip.
5. Open access code/data	Prefer Yes	Use anonymized supplementary zip or anonymous repository. Include README with exact commands. Public datasets linked and cited.
6. Experimental details	Yes	Models, features, preprocessing, hyperparameters, CV, permutations, bootstraps.
7. Statistical significance	Yes	Report CIs, permutation p-values, correlations, AUCs. Define what variability each CI captures.
8. Compute resources	Yes	MacBook Pro M4 Pro, 24GB RAM, macOS 15.3.1, CPU, approximate runtimes, any GPU runs for LLMs if applicable.
9. Code of Ethics	Yes	State you reviewed and complied. Public de-identified fMRI, no new recruitment.
10. Broader impacts	Yes or N/A with explanation	Foundational representation analysis. Discuss possible misuse/overinterpretation of brain-LLM similarity and avoid deployment claims.
11. Safeguards	N/A unless releasing high-risk model	If only analysis code and embeddings, state no high-risk pretrained model is released.
12. Licenses	Yes	Cite OpenNeuro dataset version/license, LLM/model licenses, code package licenses.
13. Assets	Yes if releasing derived data/code, otherwise N/A	If releasing processed embeddings/ROI matrices, provide documentation, license, provenance, limitations.
14. Crowdsourcing/human subjects	N/A for new collection; explain public dataset	No new participants/crowdworkers. Original human data collected by dataset creators.
15. IRB approvals	N/A for new IRB; explain	No new human-subject data collected; public de-identified dataset used. Cite original dataset documentation.
16. LLM usage declaration	Yes if LLMs are part of method	Distinguish LLMs as analysed models from any writing/code assistance. Document model use in Methods.

7. Ethics, human data, and licenses

Human data wording

Use wording like this in Methods or Ethics/Limitations:

This study uses only publicly available, de-identified neuroimaging data from the NeuroEmo dataset, OpenNeuro accession ds005700 v1.2.0. No new participants were recruited, no new human-subject data were collected, and all analyses were performed on already released BIDS-formatted data. We cite the original dataset authors and follow the access/licensing terms provided by the dataset metadata.

Why it is needed: NeurIPS reviewers and ethics reviewers need to know whether the work involved new human subjects, whether informed consent/IRB issues apply to your team, and whether the dataset is legally and ethically usable. Public fMRI data is still human-participant data, so the source, de-identification/public release status, and access terms must be explicit.

License actions before submission

Asset	Action
NeuroEmo/OpenNeuro dataset	Download dataset_description.json or dataset page metadata. Record license, version, DOI/accession, original paper citation.
LLM models	Record model names, versions/checkpoints, licenses/terms of use, source URLs.
Text emotion dataset	Record dataset source, license, version, citation.
Code libraries	Cite major libraries in methods or appendix if standard: Python, scikit-learn, Nilearn, NumPy/SciPy, PyTorch/Transformers if used.
Released derived assets	Provide provenance and an explicit license if you release processed ROI matrices, embeddings, or figures/code.

8. Compute reporting

Component	Hardware	Runtime	Notes
Brain ROI processing	Apple MacBook Pro, M4 Pro chip, 24GB RAM, macOS 15.3.1; CPU	~1-2 hours	Nilearn ROI extraction from fMRI BOLD data using Harvard-Oxford cortical atlas; approximately 40 subjects; generated 48D ROI feature matrix.
Brain decoding/statistics	Same local CPU machine	~1 hour	LOSO decoding, bootstrap confidence intervals, permutation testing, centroids, RDM, margin-uncertainty correlations, AUC, feature-shuffle control.
Data acquisition	Local machine + OpenNeuro download	~1-2+ hours, network dependent	Large public fMRI dataset download; report separately from analysis runtime.
LLM embeddings/fine-tuning	To be filled	To be filled	Add GPU/CPU type, memory, number of runs, epochs, batch size, wall-clock time.

Recommended text: All brain-data preprocessing and statistical analyses were performed locally on a MacBook Pro (Apple M4 Pro chip, 24GB RAM, macOS 15.3.1) using CPU-based Python workflows. No GPU acceleration was required for the brain-data analyses due to the relatively low-dimensional ROI representation.

9. Reproducibility package checklist

File/folder	Required content
README.md	Exact environment setup; how to download datasets; commands to reproduce all figures/tables; expected outputs.
requirements.txt or environment.yml	Package versions. Include Python version if possible.
scripts/01_prepare_brain.py	Load OpenNeuro-derived data or processed CSV; produce final 200 x 48 matrix.
scripts/02_brain_geometry.py	Centroids, RDMS, VA, decoding, margin tests.
scripts/03_llm_embeddings.py	Embedding extraction/fine-tuned model outputs.
scripts/04_llm_geometry.py	LLM centroids, margin, uncertainty, density, decay.
scripts/05_make_figures.py	Regenerate paper figures.
data/README.md	Dataset sources, links, license notes, which data are not redistributed.
results/	CSV/JSON outputs for reported numbers.
anonymization check	No usernames, local paths, author names, institution names, personal GitHub links in supplementary files.

10. Pre-upload desk-rejection checklist

Check	Pass condition
PDF format	Single PDF including paper, references, appendices, checklist in required order.
Page limit	Main content <= 9 pages including figures/tables; references and checklist excluded.

Check	Pass condition
Style file	Correct NeurIPS 2026 LaTeX style; no margin/font hacks.
Anonymity	No author names, acknowledgments, institution-specific paths, identifying GitHub links.
Supplementary zip	Anonymized; $\leq$ stated size limit; no identifying metadata.
Checklist included	Must be present; otherwise desk rejection risk.
OpenReview profiles	All authors have complete, updated profiles and conflicts.
Licenses	Dataset/model/code licenses recorded and cited.
Ethics	Human data and broader impacts addressed.
Claims	Abstract/introduction do not overclaim universality or biological equivalence.

11. Likely reviewer questions and pre-emptive answers

Question	Pre-emptive paper answer
Is this just a descriptive LLM-brain comparison?	No. The main contribution is a mechanism: uncertainty is predicted by centroid margin, validated with statistics and controls.
Why is margin meaningful?	It measures competition between the correct class and nearest alternative; unlike raw distance, it directly reflects decision boundary proximity.
Does distance alone explain uncertainty?	No. Report weak distance correlation/AUC and use it as a motivating negative result.
Is the effect random?	No. Feature shuffle preserves feature marginals but destroys multivariate structure; margin-uncertainty correlation collapses.
Does the brain dataset have confidence scores?	No. Uncertainty is estimated by cross-validated classifier probabilities; margin is validated against this independent uncertainty estimate.
Are LLMs claimed to replicate brains?	No. The claim is a shared local geometric mechanism, while global organisation can differ, e.g. valence/arousal axis differences.
Is five emotions enough?	Acknowledge as limitation; argue the contribution is a validated principle across tested systems, not a universal taxonomy of emotion.
How reproducible is this?	Provide code, seed, environment, datasets, and exact outputs.

12. Final readiness scoring

Area	Target before submission	Current action
Quality	All claims backed by stats/controls	Ensure every abstract claim maps to a result table or figure.
Clarity	A reviewer can reproduce from paper + appendix	Add formal definitions and pipeline figure.
Significance	Mechanism framed as useful beyond this dataset	Emphasize uncertainty as a general representational question.
Originality	Clear distinction from prior geometry/alignment work	State that prior work identifies geometry/alignment; this work identifies a margin-based uncertainty mechanism.
Ethics	Public human data and licenses clear	Add human-data paragraph and license table.
Reproducibility	Anonymized code/data bundle	Prepare README and run-from-scratch scripts.

Final rule: if a reviewer asks 'what exactly is new?', the answer should be one sentence: We show that representational ambiguity is governed by relative centroid competition rather than absolute distance, and validate this mechanism across LLM and brain emotion representations with statistical controls.